

PARTS LIST

Semiconductors:

- IC1 - LM324 quad operational amplifier
 IC2 - AT89C51 microcontroller
 IC3 - L293D motor driver
 T1, T2 - L14F1 photo-transistor
 D1 - 1N4007 diode
 LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R5 - 10-kilo-ohm
 R3, R4 - 5.6-kilo-ohm
 R6 - 330-ohm
 R7 - 220-ohm
 R8 - 1-kilo-ohm
 VR1, VR2 - 10-kilo-ohm preset

Capacitors:

- C1 - 10 μ F, 16V electrolytic
 C2, C3 - 33pF ceramic disk
 C4 - 47 μ F, 16V electrolytic
 C5 - 0.1 μ F ceramic disk

Miscellaneous:

- S1 - On/off switch
 S2 - Push-to-on switch
 X_{cr} - 12MHz crystal
 M1, M2 - 20 rpm, 6V DC geared motor
 Batt. - 6V, 4.5AH battery
 - Two side brackets for mounting motors
 - One castor wheel (for front wheel)
 - Two wheels for the rear
 - Chassis

ler is programmed to make the robot move forward, turn right or turn left based on the input coming from the comparator. The outputs of the microcontroller are fed to the motor driver.

Motor driver. The current supplied by the microcontroller to drive the motor is small. Therefore a motor-driver IC is used. It provides sufficient current to drive the motor.

Fig. 2 shows the circuit of the automated line-following robot. When light falls on the phototransistor (say, T1), it goes into saturation and starts conducting. When no light falls on the phototransistor, it is cut-off. A white LED (LED2) has been used to illuminate the white path on a black background. Phototransistors T1 and T2 are used for detecting the white path on the black background.

Collectors of phototransistors T1 and T2 are connected to the inverting inputs of operational amplifiers A2 and A1. The signal voltage at the inverting input of the operational amplifier is compared with the fixed reference voltage, which is formed by a potential divider circuit of 5.6-kilo-ohm resistor

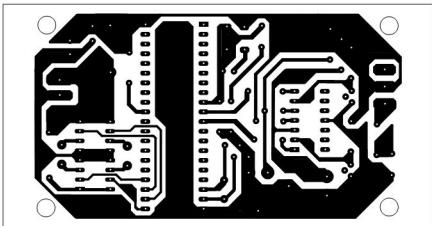


Fig. 3: Actual-size, single-side PCB for the automated line-following robot

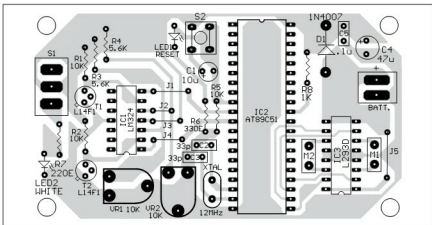


Fig. 4: Component layout for the PCB

Action Corresponding to the Microcontroller Outputs

Inputs			Outputs			Action
P3.0	P3.1	P2.3	P2.2	P2.1	P2.0	
0	0	1	0	1	0	Forward
0	1	0	0	1	0	Left
1	0	1	0	0	0	Right
1	1	0	0	0	0	Stop

and 10-kilo-ohm preset. This reference voltage can be adjusted by changing the value of the 10-kilo-ohm preset.

When sensor T2 is above the black surface, it remains cut-off as the black surface absorbs virtually all the light falling from LED2 and no light is reflected back. The voltage at the inverting input (pin 2) of operational amplifier A1 is higher than the reference voltage at its non-inverting input (pin 3) and therefore the amplifier output at

pin 1 becomes zero.

When sensor T2 is above the white line, the light gets reflected from the white surface to fall on phototransistor T2. Phototransistor T2 goes into saturation and conducts. The inverting input (pin 2) of operational amplifier

A1 goes below the reference voltage at its non-inverting input (pin 3) of operational amplifier A1 and therefore output pin 1 goes high. This way, the comparator outputs logic '0' for black surface and logic '1' for white surface.

Similarly, comparator A2 compares the input voltage from phototransistor T1 with a fixed reference voltage.

The outputs of operational amplifiers A1 and A2 are fed to microcontroller AT89C51. The AT89C51 is an 8-bit